Ring Theory

~NPTEL

[Lt-1]

• Ring Theory • ~ by Krishna Hanumanthu

Introduction to Rings and Fields

• Start with some examples of rings:

(1) $\mathbb{Z}$ = set of all integers.

There are two operations +: addition
×: multiplication

$(\mathbb{Z}, +)$ is an abelian group. ✓

$(\mathbb{Z}, \times)$ is not a group (This is not necessary for a ring.)

(2) $\mathbb{Q}$: set of all rational numbers.

It has +, ×

$(\mathbb{Q}, +)$ abelian group

$(\mathbb{Q}, \times)$ is not a group. But $\mathbb{Q}^*$ is a group under ×, i.e. $(\mathbb{Q}^*, \times)$ is an abelian group..

Similarly, we have
$\mathbb{R}$ – the set of real numbers
$\mathbb{C}$ – the set of all complex no.'s.

(3) Consider $\mathbb{Z}[i]$, where i is a complex square root of -1. ∴ $\mathbb{Z}[i] = \{a+ib \mid a, b \in \mathbb{Z}\}$

$n \in \mathbb{Z}[i]$ $\forall n \in \mathbb{Z}$, such as $1 \in \mathbb{Z}[i]$,

$3+2i \in \mathbb{Z}[i]$. Note: $\mathbb{Z}[i] \subseteq \mathbb{C}$ (subset)

We can add two elements of $\mathbb{Z}[i]$:

$$\underbrace{(a+ib)}_{\in \mathbb{Z}[i]} + \underbrace{(c+id)}_{\in \mathbb{Z}[i]} = \underbrace{(a+c)+i(b+d)}_{\in \mathbb{Z}[i]}$$

$\mathbb{Z}[i]$ is closed under addition.

It is easy to verify that $\mathbb{Z}[i]$ is an abelian group under (+). It is a subgroup of $(\mathbb{C}, +)$ (I)

Let us choose $a+ib,\ c+id \in \mathbb{Z}[i]$.

Now, $(a+ib)(c+id) = (ac-bd) + i(bc+ad) \in \mathbb{Z}[i]$

Clearly, $\mathbb{Z}[i]$ is closed under $\times$ (multiplication)

$1 \in \mathbb{Z}[i]$; it is associative. ← ①

(4) $A = \{a + \frac{b}{2} \mid a, b \in \mathbb{Z}\}$ (here we replace i by $\frac{1}{2}$)

$\frac{1}{2} \in A$ but $\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{4} \notin A$.

Ex: $\frac{1}{4}$ cannot be written as $a + \frac{b}{2}$ for some integers a and b.

$\therefore$ A is not closed under multiplication

Definition: A ring R is a set with two operations denoted by $+$ and $\times$ (called addition and multiplication) satisfying the following properties or axioms:

(1) $(R, +)$ is an abelian group.

(2) Multiplication is 'commutative' $[a, b \in R$, $a \times b = a \cdot b = ab = ba]$, associative, contains an identity element (denoted by 1).

(3) $+$ and $\times$ 'distribute': $\forall\, a, b, c \in R$,

$(a+b)c = ac + bc$

Examples: $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}[i]$ are rings.

Distributive property holds for $\mathbb{C}$, so it also holds for other sets above.

$$\begin{cases} \mathbb{Z} \subseteq \mathbb{Z}[i] \subseteq \mathbb{C} \\ \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C} \end{cases}$$

Definition: Let R be a ring. A subset S of R is a "subring" if it is closed under addition and multiplication, it is a subgroup of $(R, +)$ and contains 1.

Eg: $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{Z}[i]$ are subrings of $\mathbb{C}$.

[Lt-2]

Remark: 1. It is possible to define rings without asking for multiplication to be commutative. They are called "non-commutative rings".

Important examples: matrix rings.

Eg: 3×3 square matrix with real entries.

In this course, we will only consider commutative ring.

2. Rings can also be defined by not asking for a multiplicative identity.

Eg: $R = 2\mathbb{Z} = \{$set of all even integers$\}$ does not have multiplicative identity "1"

In this course, we will always assume rings have 1.

More examples of rings:

(1.) Zero ring: $R = \{0\}$. In this ring $1 \left(\begin{smallmatrix}\text{multiplicative}\\ \text{identity}\end{smallmatrix}\right) = 0$.

(2.) $\mathbb{Z}$, $3\mathbb{Z}$ is a subgroup of $(\mathbb{Z}, +)$

Consider the quotient group $\mathbb{Z}/3\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}\}$, where $\bar{0} = 0+3\mathbb{Z}$, $\bar{1} = 1+3\mathbb{Z}$, $\bar{2} = 2+3\mathbb{Z}$

$\mathbb{Z}/3\mathbb{Z}$ has addition $+$, and it is an abelian group.

There is also a multiplication on $\mathbb{Z}/3\mathbb{Z}$:

$\bar{0}\cdot\bar{1} = \bar{0}$, $\bar{0}\cdot\bar{2} = \bar{0}$, $\bar{1}\cdot\bar{2} = \bar{2}$

$\bar{1}\cdot\bar{0} = \bar{0}$, $\bar{2}\cdot\bar{0} = \bar{0}$, $\bar{2}\cdot\bar{1} = \bar{2}$

$\bar{0}\cdot\bar{0} = \bar{0}$, $\bar{1}\cdot\bar{1} = \bar{1}$, $\bar{2}\cdot\bar{2} = \bar{1}$

There is an identity element: 1
'×' is associative: ✓
'×' is commutative: ✓

So, $\boxed{\mathbb{Z}/3\mathbb{Z} \text{ is a ring.}}$

More generally, for every $n \geq 0$, $\mathbb{Z}/n\mathbb{Z}$ is a ring.

If $n > 0$, then the number of elements of $\mathbb{Z}/n\mathbb{Z}$ is exactly n.

Remark: For $n \geq 2$, $\mathbb{Z}/n\mathbb{Z}$ is not a subring of $\mathbb{C}$.

(3.) $R = \{f : \mathbb{R} \to \mathbb{R} \mid f \text{ is continuous}\}$

There is a ring structure on R.

'+': $f, g \in R$ i.e. $f : \mathbb{R} \to \mathbb{R}$ is continuous
$g : \mathbb{R} \to \mathbb{R}$ is continuous.

To show, $\boxed{f+g \in R}$

$f+g : \mathbb{R} \to \mathbb{R}$

in $\mathbb{R}$.

$(f+g)(a) = \boxed{f(a)+g(a)}$ for any $a \in \mathbb{R}$.

Ex: $f+g$ is continuous. So $f+g \in R$.

Is R an abelian group under $+$?

Zero element: $0 : \mathbb{R} \to \mathbb{R}$ (zero function)
is the zero function $\quad a \mapsto 0 \quad \forall a \in \mathbb{R}$.

$$(f+0)(a) = f(a) + 0(a) = f(a)$$

$$f + (-f) = 0 \Rightarrow \underbrace{(-f)(a)}_{\text{continuous function}} = -f(a)$$

∴ + is associative ✓
+ is commutative ✓

} Both follows because '+' has these properties on $\mathbb{R}$

Multiplication on $\mathbb{R}$: $f, g \in \mathbb{R}$

$$fg: \mathbb{R} \to \mathbb{R}$$
$$(fg)(a) = f(a)\, g(a)$$

Ex: • (fg) is continuous; so $fg \in R$.

• The constant function $1: \mathbb{R} \to \mathbb{R}$, $a \mapsto 1$ is the multiplicative identity.

• '$\times$' is commutative and associative

Hence, $\boxed{R \text{ is a ring.}}$

Def: Let R be a ring. An element $a \in R$ is called a "unit" if it has a multiplicative inverse.

If R is a ring (assume that R is not the zero ring $\{0\}$)

Then, 1 is a unit since, $1 \cdot 1 = 1$.

Examples: What are the units in $\mathbb{Z}$?

They are: $1, -1$ $\qquad (-1)\underbrace{(-1)}_{\text{inverse of } -1} = 1$

There are no other units in $\mathbb{Z}$:

Let $n \in \mathbb{Z}$, $n \neq 1$, $n \neq -1$

$n \boxed{?} = 1$ (not possible)

So, the only units in $\mathbb{Z}$ are 1 and -1.

Units in $\mathbb{Q}$: Every non-zero rational number is a unit in $\mathbb{Q}$.

Proof: $\frac{a}{b} \in \mathbb{Q}$, $a, b \in \mathbb{Z}$, $a \neq 0$ $b \neq 0$

$\therefore \left(\frac{a}{b}\right)\left(\frac{b}{a}\right) = 1$. So, $\left(\frac{a}{b}\right)$ has a multiplicative inverse. So it is a unit.

The set of units in $\mathbb{Q} = \mathbb{Q} - \{0\}$

(Remark: 0 is not a unit)

Def: A field in which every non-zero element is a unit.

Examples: $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are fields.

$\mathbb{Z}$ is not a field.